Dataset Description and Data Source

1. Dataset and Source

This project uses the PlantVillage dataset, introduced by Mohanty, Hughes and Salathé (2016) to support smartphone-based crop disease diagnosis. The copy used is hosted on Kaggle.

Dataset link: https://www.kaggle.com/datasets/mohitsingh1804/plantvillage

2. Dataset Content

The dataset contains 54,306 images of healthy and infected plant leaves (the Kaggle copy lists around 54,303), divided into 38 classes. The images cover 14 crop species, including tomato, potato, corn, apple and grape. Each class combines a crop with either a disease label or a healthy label. In total, 26 diseases are represented, caused by viral, bacterial, fungal, mold and mite-related conditions.

3. Image Characteristics

All images were captured in a laboratory. Each image shows a single leaf against a plain gray or black background, not a field setting. The dataset is available in three versions: original colour, grayscale, and segmented (the leaf is separated from the background). This project uses the original colour version, because most of the comparable studies used it.

4. Reason for Selection

PlantVillage is the standard dataset used by almost all of the studies cited in the literature review, so the results of this project can be compared directly with theirs. It is also large enough to fine-tune both models properly without overfitting, yet small enough to train on Google Colab without practical difficulty.

5. Data Split

The data will be divided into training, validation and test sets in an 80/10/10 ratio. The split will be stratified, so that class proportions stay the same in all three sets and no disease class is underrepresented in testing.

6. Limitation

Because all images were taken in a laboratory, the results will not show how the models perform on field photographs with complex backgrounds and lighting. This project is scoped to laboratory-condition data, and this is acknowledged as a limitation rather than addressed here.

Reference

Mohanty, S. P., Hughes, D. P., and Salathé, M. (2016). Using deep learning for image-based plant disease detection. Frontiers in Plant Science, 7, 1419.